Matrices & Matrix Algebra

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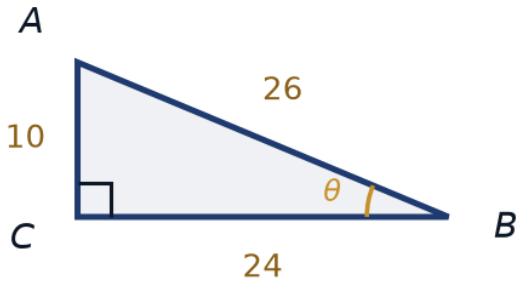
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the standard matrix $[T]$ of the linear operator $T:\mathbb{R}^2\to\mathbb{R}^2$ that performs a horizontal shear by factor 2, followed by a counterclockwise rotation of 45°.

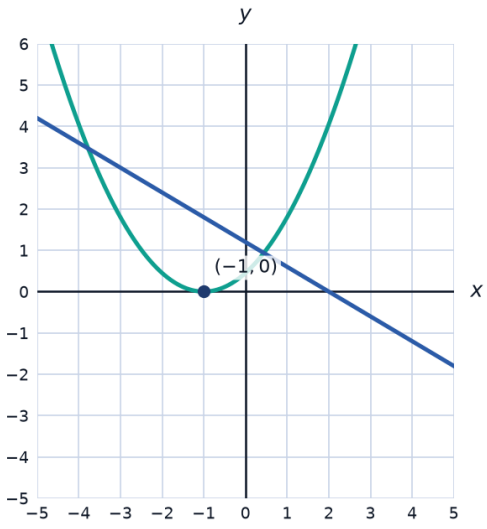
2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $T:\mathbb{R}^2\to\mathbb{R}^2$ have standard matrix $A=(4,1;2,3)$. Find the matrix representation $[T]_{\mathcal{B}}$ relative to the basis $\mathcal{B}=\{(1;1),(-1;2)\}$ via similarity transformation $P^{-1}AP$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

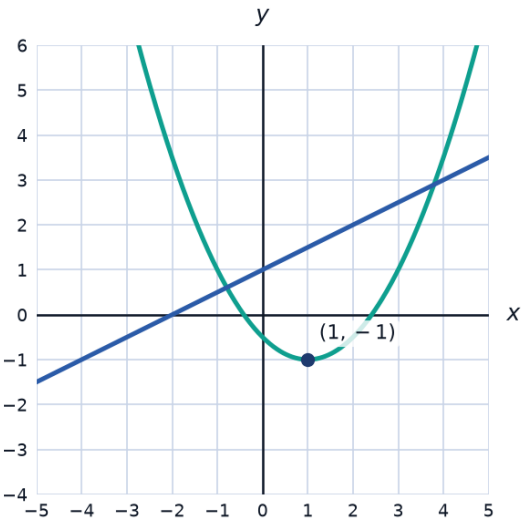
Use the Gauss–Jordan elimination algorithm on the augmented block matrix $[A\mid I_3]$ to compute A^{-1} for: $A=(1,0,2;2,-1,3;4,1,8)$.



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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Prove that if λ is an eigenvalue of an invertible matrix A, then $\lambda eq 0$ and λ^{-1} is an eigenvalue of A^{-1} with the exact same corresponding eigenvector.

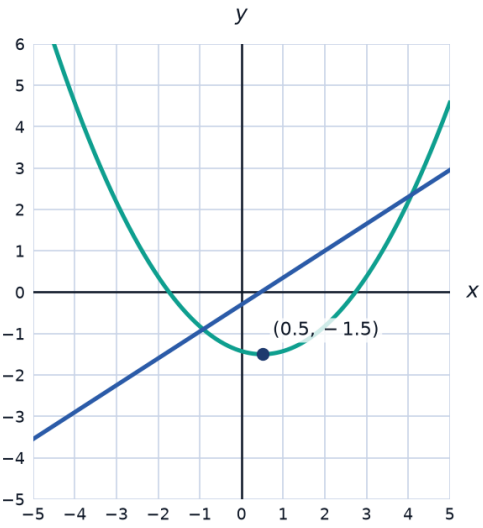


5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Diagonalize the matrix $A = (1, 2; 2, 1)$ by finding invertible P and diagonal D such that $A = PDP^{-1}$. Then compute A^k for any positive integer k.

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the matrix $A = (1, 2, 0, 1; 2, 4, 1, 4; 3, 6, 1, 5)$, find bases for $\text{Row}(A)$, $\text{Col}(A)$, and $\text{Null}(A)$. Verify the Rank–Nullity Theorem.



7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

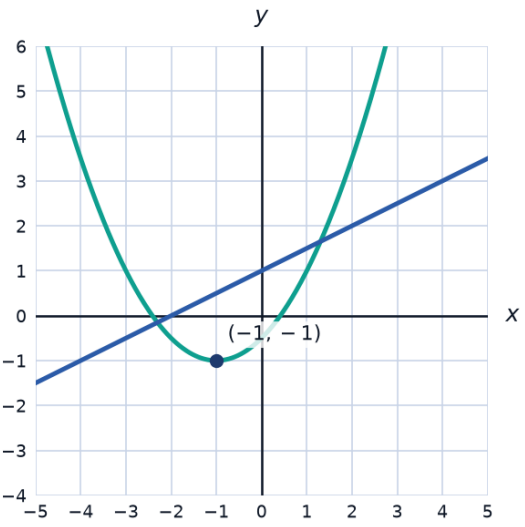
Consider a 2×2 matrix A with an eigenvalue $\lambda = 3$ and a corresponding eigenvector $\mathbf{v} = (1; 2)$. If the trace of A is 7, find the other eigenvalue of A.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Let $A = (1, 2, 0; 0, 3, 1; 2, 1, 4)$. Compute $\det(A)$, find the adjugate matrix $\operatorname{adj}(A)$, and verify the identity $A\operatorname{adj}(A) = \det(A)I_3$.



9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Let A and B be invertible $n \times n$ matrices. Prove algebraically that: (a) $(AB)^T = B^T A^T$, and (b) $(AB)^{-1} = B^{-1}A^{-1}$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use Cramer's Rule to solve for x_2 in the linear system: $\{2x_1 + x_2 + x_3 = 4; x_1 - x_2 + 2x_3 = 1; 3x_1 + 2x_2 - x_3 = 7$

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Compute the determinant of the 4×4 matrix by Laplace cofactor expansion along an optimal row or column: $A = (2, 0, -1, 3; 0, 5, 0, 0; 1, -2, 3, 1; 4, 0, 1, 2)$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the matrix $A = (2, 1; 1, 3)$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $(x'; y') = A(x; y)$, what is the image of the point $(3, 1)$ under this transformation?

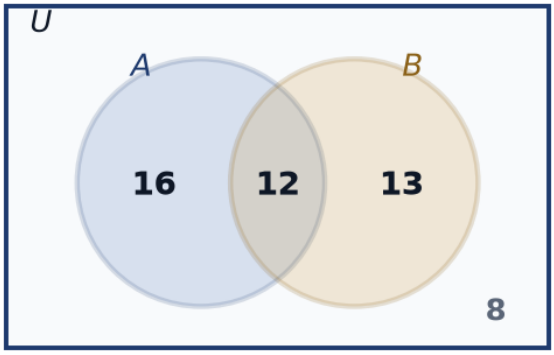
13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the component orthogonal to $\mathbf{u}$ of $\mathbf{y} = \langle -2, 3, -4 \rangle$ for $\mathbf{u} = \langle 1, -3, 3 \rangle$.

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

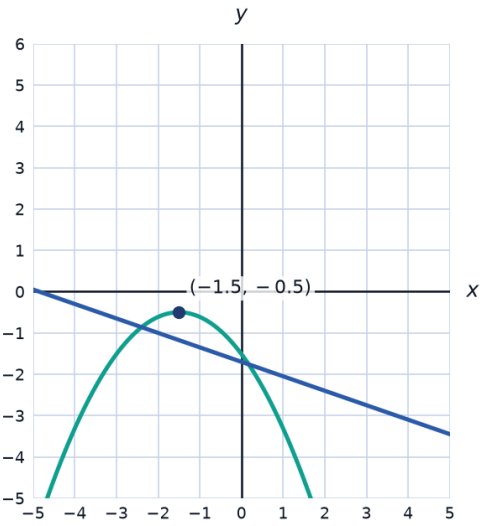
Prove that if $\mathbf{x}_p$ is any particular solution to $A\mathbf{x} = \mathbf{b}$, then the complete solution set is the affine subspace $S = \{\mathbf{x}_p + \mathbf{x}_h : \mathbf{x}_h \in \text{Null}(A)\}$.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Systems of Linear Equations

Find the shortest Euclidean distance from the point $P(3, -1, 4)$ to the hyperplane $\mathcal{H} : 2x - y + 2z = 5$ in $\mathbb{R}^3$.



16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Determinants

Find all eigenvalues, their algebraic and geometric multiplicities, and a basis for each eigenspace for: $A = (3, 1, 0; 0, 3, 0; 0, 0, 2)$. Is A diagonalizable?

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17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Vectors in n Dimensions

Apply the Gram–Schmidt orthogonalization process to the columns of A to find its QR factorization: $A = (1, 2; 2, 1; -2, 2)$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Systems of Linear Equations

Perform a complete bifurcation analysis of the system in terms of parameter $k \in \mathbb{R}$: $\{x + y + kz = 2; 3x + 4y + 2z = k; 2x + 3y - z = 1$ Determine all values of k for which the system has (a) a unique solution, (b) infinitely many solutions, and (c) no solution.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Determinants

Level 3 · Advanced / University RigorSuppose an $n \times n$ matrix A satisfies the polynomial equation $A^3 - 4A^2 + 2A - 5I = 0$. Prove that A is invertible, and express A^{-1} as an explicit polynomial in A.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Vectors in n Dimensions

Level 2 · MediumLet $A = (3, 5; 1, 2)$ and $\mathbf{b} = (4; 1)$. Compute A^{-1} and use it to solve the linear system $A\mathbf{x} = \mathbf{b}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If matrix A has size 3×4 and matrix B has size 4×2, what is the size of the product AB?

a. 3×2

b. 4×4

c. 2×3

d. Multiplication is undefined

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Is matrix multiplication commutative in general ($AB = BA$)?

a. No, $AB \neq BA$ in general even when both products are defined

b. Yes, multiplication of numbers and matrices is always commutative

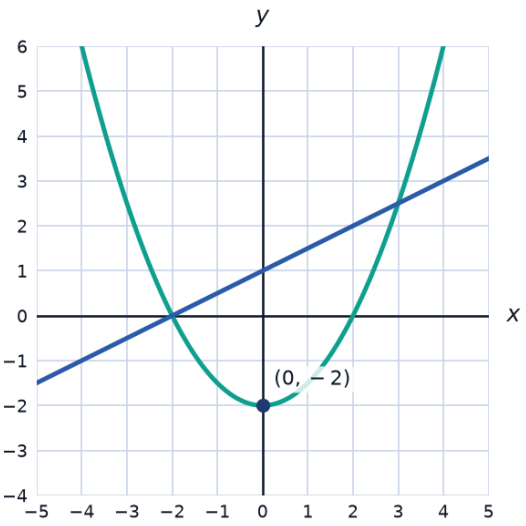
c. Only if the entries are integers

d. Yes, provided both matrices are square

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is $(AB)^T$ equal to in terms of the transposes A^T and B^T?



a. B^TA^T

b. A^TB^T

c. AB^T

d. BA^T

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the inverse of the 2×2 matrix $A = [[4, 7], [1, 2]]$.

a. $[[2, -7], [-1, 4]]$

b. $[[-2, 7], [1, -4]]$

c. $[[4, -7], [-1, 2]]$

d. $[[2, 1], [7, 4]]$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is an elementary matrix?

a. A matrix obtained by performing a single elementary row operation on an identity matrix

b. Any matrix containing only 0s and 1s

c. A diagonal matrix with determinant 1

d. A matrix with exactly one non-zero row

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If A, B, C are invertible square matrices, what is $(ABC)^{-1}$?

a. $C^{-1}B^{-1}A^{-1}$

b. $A^{-1}B^{-1}C^{-1}$

c. CBA

d. ABC^{-1}

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A square matrix A is symmetric if and only if:

a. $A^T = A$

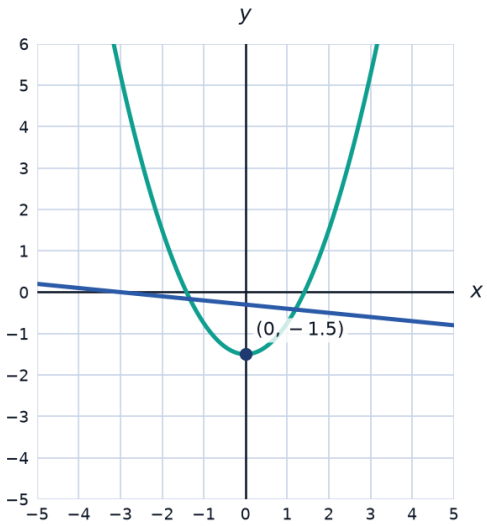
b. $A^T = -A$

c. $A^T = A^{-1}$

d. $\det(A) = 1$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What defines an orthogonal matrix Q?



a. $Q^TQ = I$, meaning $Q^{-1} = Q^T$

b. $Q^2 = I$

c. $\det(Q) = 0$

d. All entries of Q are perpendicular

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If Q is an orthogonal matrix, what are the possible values for $\det(Q)$?

a. ± 1

b. 0

c. 1 only

d. Any real number

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear if and only if for all vectors $\mathbf{u}$, $\mathbf{v}$ and scalars c:

a. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$

b. $T(\mathbf{u} \cdot \mathbf{v}) = T(\mathbf{u}) \cdot T(\mathbf{v})$

c. $T(\mathbf{u}) = ||\mathbf{u}||$

d. $T(\mathbf{0}) = [1, 1, \ldots, 1]^T$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the standard matrix for counterclockwise rotation in $\mathbb{R}^2$ by angle θ?

a. $[[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$

b. $[[\cos\theta, \sin\theta], [-\sin\theta, \cos\theta]]$

c. $[[\sin\theta, \cos\theta], [\cos\theta, -\sin\theta]]$

d. $[[\cos\theta, 0], [0, \sin\theta]]$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the standard matrix for reflection across the x-axis in $\mathbb{R}^2$?

a. $[[1, 0], [0, -1]]$

b. $[[-1, 0], [0, 1]]$

c. $[[0, 1], [1, 0]]$

d. $[[-1, 0], [0, -1]]$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the matrix $A = (2, 1; 3, 4)$. What is the matrix A^T?

a. $(2, 3; 1, 4)$

b. $(1, 2; 4, 3)$

c. $(4, 1; 3, 2)$

d. $(2, 4; 1, 3)$

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34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Consider the matrix $A = (2, 0; 0, 3)$. If $\mathbf{v} = (1; 2)$, what is $A\mathbf{v}$?

a. $(2; 3)$

b. $(1; 6)$

c. $(2; 6)$

d. $(3; 2)$

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Which matrix is orthogonal (satisfies $Q^TQ = I$)?

a. $[(1, 1; 0, 1)]$

b. $\left[\left(\frac{1}{2}, \frac{1}{2}; \frac{1}{2}, -\frac{1}{2}\right)\right]$

c. $[(0, -1; 1, 0)]$

d. $[(1, -1; 1, 1)]$

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A 3×3 matrix has eigenvalues $-5, 2, 2$. Find its determinant.

a. -20

b. -1

c. -19

d. 20

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A 2×2 matrix has eigenvalues $-1, -5$. Find its trace.

a. -6

b. 5

c. -5

d. 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Systems of Linear Equations

A parallelogram in the xy-plane is defined by the vertices $(0, 0)$, $(3, 1)$, $(1, 4)$, and $(4, 5)$. What is the area of this parallelogram?

a. 11

b. 13

c. 15

d. 17

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Determinants

A linear map T satisfies $T\langle -1, -1\rangle = \langle 0, -2\rangle$ and $T\langle 0, -1\rangle = \langle 2, -1\rangle$. Find $T\langle 2, -3\rangle$.

a. $\langle -6, -1\rangle$

b. $\langle -10, 1\rangle$

c. $\langle 10, -1\rangle$

d. $\langle 11, -1\rangle$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Vectors in n Dimensions

Find the image of $\langle 1, 1\rangle$ under the rotation counterclockwise by 30°.

a. $\langle -\frac{1}{2} + \frac{\sqrt{3}}{2}, \frac{1}{2} + \frac{\sqrt{3}}{2}\rangle$

b. $\langle \frac{1}{2} + \frac{\sqrt{3}}{2}, -\frac{1}{2} + \frac{\sqrt{3}}{2}\rangle$

c. $\langle \frac{1}{2} - \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2} - \frac{1}{2}\rangle$

d. $\langle 1, 1\rangle$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Systems of Linear Equations

Which set is NOT orthonormal?

a. $\{\langle \frac{5}{13}, \frac{12}{13}\rangle, \langle \frac{12}{13}, -\frac{5}{13}\rangle\}$

b. $\{\langle \frac{1}{2}, \frac{1}{2}\rangle, \langle \frac{1}{2}, -\frac{1}{2}\rangle\}$

c. $\{\langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3}\rangle, \langle \frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\rangle\}$

d. $\{\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\rangle, \langle \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\rangle\}$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Determinants

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 1, -1 \rangle$, $\mathbf{x}_2 = \langle 2, 3 \rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

a. $\langle 2, 3 \rangle$

b. $\langle \frac{3}{2}, \frac{7}{2} \rangle$

c. $\langle 1, 1 \rangle$

d. $\langle 1, -1 \rangle$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Vectors in n Dimensions

Find the least-squares line for the points $(-1, 3), (1, -3), (3, 1), (4, 1)$.

a. $y = \frac{47}{59} - \frac{10x}{59}$

b. $y = \frac{47x}{59} - \frac{10}{59}$

c. $y = \frac{106}{59} - \frac{10x}{59}$

d. $y = \frac{10x}{59} + \frac{47}{59}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Systems of Linear Equations

Which vector spans the kernel of $T(\mathbf{x}) = [(-2, 3, 0; 3, -1, 1)]\mathbf{x}$?

a. $\langle -2, -3, 7 \rangle$

b. $\langle -3, 2, 7 \rangle$

c. $\langle -3, -2, 7 \rangle$

d. $\langle 1, 1, 1 \rangle$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Determinants

Find the y-intercept of the least-squares line for $(-2, -2), (1, -3), (3, 6), (4, -3)$.

a. $\frac{10}{21}$

b. $-\frac{17}{14}$

c. $-\frac{3}{14}$

d. $\frac{17}{14}$

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